Additional Exercises

22 September 2025

Limits and Derivatives

1. Find numbers a and b such that

$$\lim_{x\to 0} \frac{\sqrt{ax+b}-2}{x} = 1.$$

Observe that as x approaches 0,

$$\frac{\sqrt{ax+b}-2}{x} \to \frac{\sqrt{b}-2}{0}.$$

Since the limit exists, we require that $\sqrt{b}-2=0$, or $b=4$, and we have

$$\lim_{x\to 0} \frac{\sqrt{ax+4}-2}{x} = 1.$$

We rationalize the numerator:

$$\frac{\sqrt{ax+4}-2}{x} \cdot \frac{\sqrt{ax+4}+2}{\sqrt{ax+4}+2} = \frac{ax}{x\left(\sqrt{ax+4}+2\right)} = \frac{a}{\sqrt{ax+4}+2}.$$

Thus, we have

$$\lim_{x\to 0} \frac{\sqrt{ax+4}-2}{x} = \lim_{x\to 0} \frac{a}{\sqrt{ax+4}+2} = \frac{a}{\sqrt{4}+2} = \frac{a}{4} = 1 \quad \Longrightarrow \quad a = 4.$$

2. Consider the graph of the function

$$y = x^4 - 2x^2 - x.$$

Find the two points on the curve that have a common tangent line. You do *not* need to verify that there are only two such points.

The slope of the tangent line at any point is given by $y' = 4y^3 - 4y - 1$. When $x = a$, the line passes through the point $\left(a, a^4 - 2a^2 - a\right)$, so the equation of the tangent line is given by

$$4a^3 - 4a - 1 = \frac{y - \left(a^4 - 2a^2 - a\right)}{x - a}.$$

Simplifying this, we have

$$\begin{aligned} \left(4a^3 - 4a - 1\right)x - 4a^4 + 4a^2 + a &= y - a^4 + 2a^2 + a \\ \left(4a^3 - 4a - 1\right)x - 3a^4 + 2a^2 &= y. \end{aligned}$$

Since there are two points, say $x = a$ and $x = b$ with $a \neq b$, that have the same tangent line, we have

$$y = \left(4a^3 - 4a - 1\right)x - 3a^4 + 2a^2, \quad y = \left(4b^3 - 4b - 1\right)x - 3b^4 + 2b^2.$$

Comparing coefficients, we require that

$$4a^3 - 4a - 1 = 4b^3 - 4b - 1, \quad -3a^4 + 2a^2 = -3b^4 + 2b^2.$$

From the first equation, we have

$$a^3 - a = b^3 - b \quad \Longrightarrow \quad \frac{a^3 - b^3}{a - b} = 1 \quad \Longrightarrow \quad a^2 + ab + b^2 = 1.$$

From the second equation, we have

$$3a^4 - 3b^4 = 2a^2 - 2b^2 \quad \Longrightarrow \quad 3\left(a^2 + b^2\right)\left(a^2 - b^2\right) = 2\left(a^2 - b^2\right).$$

This equality holds when $a = \pm b$. Since $a \neq b$, we have $a = -b$: plugging this into the first equation yields

$$a^2 - a^2 + a^2 = 1 \quad \Longrightarrow \quad a = \pm 1.$$

Thus, we find that $a = \pm 1$ and $b = \mp 1$, and so the two points with the common tangent line are $(1, -2)$ and $(-1, 0)$.

Integration

1. (a) Use an appropriate substitution to show that, for any continuous function f,

$$\int_0^a f(x)\, dx = \int_0^a f(a - x)\, dx.$$

(b) Hence, or otherwise, show that

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x}\, dx = \frac{\pi}{4}.$$

(a) Let $u = a - x$, with $du = -dx$. Then,

$$\int_0^a f(a - x)\, dx = \int_{a-0}^{a-a} f(u) \cdot -1\, du = -\int_a^0 f(u)\, du = \int_0^a f(u)\, du.$$

This is precisely the desired result.

(b) Directly applying the previous result, we have

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x}\, dx = \int_0^{\pi/2} \frac{\sin^3(\pi/2 - x)}{\sin^3(\pi/2 - x) + \cos^3(\pi/2 - x)}\, dx.$$

Since $\sin(\pi/2 - x) = \cos x$ and $\cos(\pi/2 - x) = \sin x$, we may rewrite this integral as

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x}\, dx = \int_0^{\pi/2} \frac{\cos^3 x}{\cos^3 x + \sin^3 x}\, dx.$$

Observe, then, that

$$\begin{aligned} 2\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} &= \int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x}\, dx + \int_0^{\pi/2} \frac{\cos^3 x}{\cos^3 x + \sin^3 x}\, dx \\ &= \int_0^{\pi/2} \frac{\sin^3 x + \cos^3 x}{\sin^3 x + \cos^3 x}\, dx = \int_0^{\pi/2} 1\, dx = \frac{\pi}{2}. \end{aligned}$$

Hence, we conclude

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin^3 x + \cos^3 x} = \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}.$$

2. Evaluate the integral
$$\int \frac{1}{x^7 - x}\, dx.$$
Hint: Try a substitution!

We divide the integrand throughout by the highest power:
$$\int \frac{x^{-7}}{1 - x^{-6}}\, dx.$$
Let $u = 1 - x^{-6}$, with $du = 6x^{-7}\, dx$. Then,
$$\int \frac{1/6}{u}\, du = \frac{1}{6} \ln|u| + c = \frac{1}{6} \ln\left|1 - x^{-6}\right| + c.$$

3. Use integration by parts to evaluate
$$\int e^x \sin x\, dx.$$
Hint: You might need to use integration by parts twice.

We integrate by parts: let $u = \sin x$ and $dv = e^x\, dx$, with $du = \cos x\, dx$ and $v = e^x$. Then,
$$\int e^x \sin x\, dx = e^x \sin x - \int e^x \cos x\, dx.$$
To evaluate the integral of $e^x \cos x$, we integrate by parts again: let $u = \cos x$ and $dv = e^x\, dx$, with $du = -\sin x\, dx$ and $v = e^x$. Then,
$$\begin{aligned}\int e^x \sin x\, dx &= e^x \sin x - \left[e^x \cos x - \int e^x\,(-\sin x)\, dx\right] \\ &= e^x \sin x - e^x \cos x - \int e^x \sin x\, dx.\end{aligned}$$
Rearranging terms, we find that
$$\int e^x \sin x\, dx = \frac{1}{2}\left(e^x \sin x - e^x \cos x\right) + c.$$

4. A solid is formed by rotating the the graph of the curve
$$9x = y^2 + 18, \quad 2 \le x \le 6$$
about the x-axis.

(a) Find the exact volume of the solid.

(b) Find the exact surface area of the solid.

Since the curve is symmetric about the x-axis, the same solid is formed by rotating $y = \sqrt{9x - 18}$ about the x-axis.

(a) The volume is given by
$$\begin{aligned}\int_2^6 \pi y^2\, dx = \pi \int_2^6 9x - 18\, dx &= \pi \left[\frac{9}{2}x^2 - 18x\right]_2^6 \\ &= \pi \left[\left(\frac{9}{2} \cdot 36 - 18 \cdot 6\right) - \left(\frac{9}{2} \cdot 4 - 18 \cdot 2\right)\right] = 72\pi.\end{aligned}$$

(b) The surface area is given by

$$\int_2^6 2\pi y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx.$$

Note that

$$\frac{dy}{dx}=\frac{1}{2}(9x-18)^{-1/2}\cdot 9=\frac{9}{2\sqrt{9x-18}}.$$

Hence, the surface area is

$$\int_2^6 2\pi\sqrt{9(x-2)}\sqrt{1+\frac{9}{4(x-2)}}\,dx=6\pi\int_2^6\sqrt{x+\frac{1}{4}}\,dx.$$

Letting $u=x+1/4$ with $du=dx$,

$$\begin{aligned}6\pi\int_2^6\sqrt{x+\frac{1}{4}}\,dx&=6\pi\int_{9/4}^{25/4}\sqrt{u}\,dx=6\pi\cdot\frac{2}{3}u^{3/2}\Big|_{9/4}^{25/4}\\&=4\pi\left[\left(\frac{25}{4}\right)^{3/2}-\left(\frac{9}{4}\right)^{3/2}\right]=49\pi.\end{aligned}$$

Infinite Sequences and Series

1. Find the sum of the series

$$\sum_{n=1}^{\infty}\frac{1}{n(n+3)}.$$

We peform a partial fraction decomposition on each term of the sum:

$$\frac{1}{n(n+3)}=\frac{A}{n}+\frac{B}{n+3}=\frac{(A+B)n+3A}{n(n+3)}.$$

We thus require that $A+B=0$, and $3A=1$. Hence, $A=1/3$ and $B=-1/3$, and so

$$\sum_{n=1}^{\infty}\frac{1}{n(n+3)}=\frac{1}{3}\sum_{n=1}^{\infty}\frac{1}{n}-\frac{1}{n+3}.$$

We now consider the partial sums: observe that

$$\begin{aligned}\frac{1}{3}\sum_{k=1}^{n}\frac{1}{k}-\frac{1}{k+3}=\frac{1}{3}\Bigg[&\left(\frac{1}{1}-\frac{1}{4}\right)+\left(\frac{1}{2}-\frac{1}{5}\right)+\left(\frac{1}{3}-\frac{1}{6}\right)\\&+\left(\frac{1}{4}-\frac{1}{7}\right)+\left(\frac{1}{5}-\frac{1}{8}\right)+\left(\frac{1}{6}-\frac{1}{9}\right)+\cdots\\&+\left(\frac{1}{n-5}-\frac{1}{n-2}\right)+\left(\frac{1}{n-4}-\frac{1}{n-1}\right)+\left(\frac{1}{n-3}-\frac{1}{n}\right)\\&+\left(\frac{1}{n-2}-\frac{1}{n+1}\right)+\left(\frac{1}{n-1}-\frac{1}{n+2}\right)+\left(\frac{1}{n}-\frac{1}{n+3}\right)\Bigg].\end{aligned}$$

Cancelling terms, we are left with

$$\frac{1}{3}\sum_{k=1}^{n}\frac{1}{k}-\frac{1}{k+3}=\frac{1}{3}\left[\frac{1}{1}+\frac{1}{2}+\frac{1}{3}-\frac{1}{n+1}-\frac{1}{n+2}-\frac{1}{n+3}\right].$$

As n approaches infinity, we thus find that

$$\sum_{n=1}^{\infty}\frac{1}{n(n+3)}=\lim_{n\to\infty}\frac{1}{3}\left[\frac{1}{1}+\frac{1}{2}+\frac{1}{3}-\frac{1}{n+1}-\frac{1}{n+2}-\frac{1}{n+3}\right]=\frac{11}{18}.$$

2. (a) Show that
$$n^2+n+1=\frac{1+n(n+1)}{(n+1)-n}.$$
(b) Find the sum of the series
$$\sum_{n=0}^{\infty}\cot^{-1}\left(n^2+n+1\right).$$
Hint: You may use the fact that
$$\cot^{-1}\frac{1+ab}{b-a}=\cot^{-1}a-\cot^{-1}b.$$

(a) Expanding the right-hand side,
$$\frac{1+n(n+1)}{(n+1)-n}=\frac{n^2+n+1}{1}=n^2+n+1.$$

(b) We may rewrite the sum as
$$\sum_{n=0}^{\infty}\cot^{-1}\left(\frac{1+n(n+1)}{(n+1)-n}\right)=\sum_{n=0}^{\infty}\cot^{-1}n-\cot^{-1}(n+1).$$
We consider the partial sums:
$$\begin{aligned}\sum_{k=0}^{n}\cot^{-1}k-\cot^{-1}(k+1)&=\left(\cot^{-1}0-\cot^{-1}1\right)+\left(\cot^{-1}1-\cot^{-1}2\right)+\cdots+\\&\quad\left(\cot^{-1}(n-1)-\cot^{-1}n\right)+\left(\cot^{-1}n-\cot^{-1}(n+1)\right)\\&=\cot^{-1}0-\cot^{-1}(n+1).\end{aligned}$$
Thus, as n approaches infinity,
$$\sum_{n=0}^{\infty}\cot^{-1}\left(n^2+n+1\right)=\lim_{n\to\infty}\left[\cot^{-1}0-\cot^{-1}(n+1)\right]=\frac{\pi}{2}.$$

3. Let $f(x)=\sin\left(x^3\right)$.
(a) Write down the Maclaurin series expansion of $f(x)=\sin\left(x^3\right)$.
(b) Hence, or otherwise, find $f^{(15)}(0)$.

(a) The Maclaurin series expansion is given by
$$\sin\left(x^3\right)=\sum_{n=0}^{\infty}(-1)^n\frac{\left(x^3\right)^{2n+1}}{(2n+1)!}=\left(x^3\right)-\frac{\left(x^3\right)^3}{3!}+\frac{\left(x^3\right)^5}{5!}-\frac{\left(x^3\right)^7}{7!}+\cdots$$

(b) Note that $f^{(15)}(0)$ appears in the coefficient of x^{15} in the Maclaurin series expansion of $\sin\left(x^3\right)$:
$$\frac{f^{(15)}(0)}{15!}x^{15}=\frac{1}{5!}\left(x^3\right)^5\quad\Longrightarrow\quad f^{(15)}(0)=\frac{15!}{5!}.$$

4. Use an appropriate Maclaurin series to evaluate the sum

$$\sum_{n=0}^{\infty} \frac{(x+2)^n}{(n+3)!}.$$

Observe that

$$\begin{aligned}\sum_{n=0}^{\infty} \frac{(x+2)^n}{(n+3)!} &= \sum_{n=0}^{\infty} \frac{(x+2)^n}{(n+3)!} \cdot \frac{(x+2)^3}{(x+2)^3} \\ &= \frac{1}{(x+2)^3} \sum_{n=0}^{\infty} \frac{(x+2)^{n+3}}{(n+3)!} \\ &= \frac{1}{(x+2)^3} \sum_{n=3}^{\infty} \frac{(x+2)^n}{n!}\end{aligned}$$

The sum in the expression above is simply the Maclaurin series of e^{x+2} with the first three terms removed:

$$e^{x+2} = \sum_{n=0}^{\infty} \frac{(x+2)^n}{n!} = 1 + (x+2) + \frac{(x+2)^2}{2!} + \frac{(x+2)^3}{3!} + \cdots.$$

Hence, we find that

$$\sum_{n=0}^{\infty} \frac{(x+2)^n}{(n+3)!} = \frac{1}{(x+2)^3}\left[e^{x+2} - \left(1 + (x+2) + \frac{(x+2)^2}{2!}\right)\right].$$

Vectors and Vector Functions

1. A tetrahedron has vertices at the points $A\,(0,0,3)$, $B\left(0,\sqrt{8},-1\right)$, $C\left(-\sqrt{6},-\sqrt{2},-1\right)$, and $D\left(\sqrt{6},-\sqrt{2},-1\right)$.

   (a) Show that the tetrahedron is regular: that is, show that all of its edges have equal length.

   (b) Find the distance between two edges that do not intersect.

   (c) Find the distance from a vertex to the face directly opposite.

   (a) We show that all the sides have equal length.

$$\begin{aligned}\overline{AB} &= \sqrt{(0-0)^2 + \left(\sqrt{8}-0\right)^2 + (-1-3)^2} = \sqrt{24}, \\ \overline{AC} &= \sqrt{\left(-\sqrt{6}-0\right)^2 + \left(-\sqrt{2}-0\right)^2 + (-1-3)^2} = \sqrt{24}, \\ \overline{AD} &= \sqrt{\left(\sqrt{6}-0\right)^2 + \left(-\sqrt{2}-0\right)^2 + (-1-3)^2} = \sqrt{24}, \\ \overline{BC} &= \sqrt{\left(-\sqrt{6}-0\right)^2 + \left(-\sqrt{2}-\sqrt{8}\right)^2 + (-1+1)^2} = \sqrt{24}, \\ \overline{BD} &= \sqrt{\left(\sqrt{6}-0\right)^2 + \left(-\sqrt{2}-\sqrt{8}\right)^2 + (-1+1)^2} = \sqrt{24}, \\ \overline{CD} &= \sqrt{\left(\sqrt{6}+\sqrt{6}\right)^2 + \left(-\sqrt{2}+\sqrt{2}\right)^2 + (-1+1)^2} = \sqrt{24}.\end{aligned}$$

(b) Consider the edges $\overline{AB}$ and $\overline{CD}$: a vector perpendicular to both edges is given by

$$\vec{AB} \times \vec{CD} = \begin{pmatrix} 0 \\ \sqrt{8} \\ -4 \end{pmatrix} \times \begin{pmatrix} 2\sqrt{6} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -8\sqrt{6} \\ -8\sqrt{3} \end{pmatrix}.$$

The distance between non-intersecting edges is just the length of the projection of $\vec{AC}$ onto this normal:

$$\frac{\left|\vec{AC} \cdot \left(\vec{AB} \times \vec{CD}\right)\right|}{\left|\vec{AB} \times \vec{CD}\right|} = \frac{\left|\begin{pmatrix} -\sqrt{6} \\ -\sqrt{2} \\ -4 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ -8\sqrt{6} \\ -8\sqrt{3} \end{pmatrix}\right|}{\sqrt{0^2 + \left(8\sqrt{6}\right)^2 + \left(8\sqrt{3}\right)^2}} = 2\sqrt{3}.$$

(c) We consider the distance from A to the plane Π containing $\triangle BCD$. A normal vector to Π is given by

$$\vec{BC} \times \vec{BD} = \begin{pmatrix} -\sqrt{6} \\ -3\sqrt{2} \\ 0 \end{pmatrix} \times \begin{pmatrix} \sqrt{6} \\ -3\sqrt{2} \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 12\sqrt{3} \end{pmatrix}.$$

A normal vector is thus given by $\vec{n} = \vec{k}$. (Alternatively, one can observe that the points B, C, and D all have -1 in the third component, and so must be contained in the plane $z = -1$.) The distance between A and Π is simply the length projection of $\vec{AB}$ onto $\vec{n}$:

$$\frac{\left|\vec{AB} \cdot \vec{n}\right|}{|\vec{n}|} = \frac{|-1-3|}{1} = 4.$$

2. A curve in $\mathbb{R}^3$ is parametrized by the vector function

$$\vec{r}(t) = \left(\frac{t^3}{3} - t\right)\vec{i} + \left(t^2 - 1\right)\vec{j}, \quad -2 \le t \le 2.$$

(a) The *curvature* of the curve is given by

$$\kappa = \frac{|\vec{r}'(t) \times \vec{r}''(t)|}{|\vec{r}'(t)|^3}.$$

Find the curvature of $\vec{r}(t)$ at the point $(0, -1, 0)$.

(b) Find the arc length of the curve.

(a) Note that at the point $(0, -1, 0)$, we have $t = 0$. We compute the first and second derivatives of $\vec{r}(t)$:

$$\vec{r}'(t) = \left(t^2 - 1\right)\vec{i} + 2t\,\vec{j}, \quad \vec{r}''(t) = 2t\,\vec{i} + 2\,\vec{j}.$$

Thus,

$$\vec{r}'(0) = -\vec{i}, \quad \vec{r}''(0) = 2\,\vec{j}.$$

Hence, at the point $(0, -1, 0)$,

$$\kappa = \frac{|\vec{r}'(0) \times \vec{r}''(0)|}{|\vec{r}'(0)|^3} = \frac{\left|\begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}\right|}{\left(\sqrt{(-1)^2 + 0^2 + 0^2}\right)^3} = 2.$$

(b) The arc length is given by

$$\begin{aligned}\int_{-2}^{2} |\vec{r}'(t)|\, dt &= \int_{-2}^{2} \sqrt{(t^2-1)^2 + (2t)^2}\, dt \\ &= \int_{-2}^{2} \sqrt{t^4 + 2t^2 + 1}\, dt \\ &= \int_{-2}^{2} t^2 + 1\, dt = \left.\frac{t^3}{3} + t\right|_{-2}^{2} = \frac{28}{3}.\end{aligned}$$